

Lipophilic and hydrophilic antioxidant capacities of common foods in the United States.

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Both lipophilic and hydrophilic antioxidant capacities were determined using the oxygen radical absorbance capacity (ORAC(FL)) assay with fluorescein as the fluorescent probe and 2,2'-azobis(2-amidinopropane) dihydrochloride as a peroxy radical generator on over 100 different kinds of foods, including fruits, vegetables, nuts, dried fruits, spices, cereals, infant, and other foods. Most of the foods were collected from four different regions and during two different seasons in U.S. markets. Total phenolics of each sample were also measured using the Folin-Ciocalteu reagent. Hydrophilic ORAC(FL) values (H-ORAC(FL)) ranged from 0.87 to 2641 micromol of Trolox equivalents (TE)/g among all of the foods, whereas lipophilic ORAC(FL) values (L-ORAC(FL)) ranged from 0.07 to 1611 micromol of TE/g. Generally, L-ORAC(FL) values were <10% of the H-ORAC(FL) values except for a very few samples. Total antioxidant capacity was calculated by combining L-ORAC(FL) and H-ORAC(FL). Differences of ORAC(FL) values in fruits and vegetables from different seasons and regions were relatively large for some foods but could not be analyzed in detail because of the sampling scheme. Two different processing methods, cooking and peeling, were used on selected foods to evaluate the impact of processing on ORAC(FL). The data demonstrated that processing can have significant effects on ORAC(FL). Considering all of the foods analyzed, the relationship between TP and H-ORAC(FL) showed a very weak correlation. Total hydrophilic and lipophilic antioxidant capacity intakes were calculated to be 5558 and 166 micromol of TE/day, respectively, on the basis of data from the USDA Continuing Survey of Food Intakes by Individuals (1994-1996).